United States Patent	12397871
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Rea; Eric

Three-wheel motor vehicle and control system

Abstract

A three-wheeled vehicle having a front wheel assembly attached to a chassis. The chassis includes a rotational control shaft having a rotational axis that is generally directed in a longitudinal direction of the vehicle. The rotational control shaft is integrated with or secured to the chassis in a non-rotational manner and passes through the front wheel assembly in a rotationally-free manner, such that the rotational control shaft can rotate about its rotational axis. The front wheel assembly includes one or more lean control motors, which are operably configured to rotate the rotational control shaft about its rotational axis thereby causing the chassis to lean from side to side to improve the handling ability of the vehicle. Some embodiments include a lean control system configured to automatically control the degree of rotation of the chassis.

Inventors: **Rea; Eric (Seffner, FL)**

Applicant: **Rea; Eric** (Seffner, FL)

Family ID: **1000008781166**

Appl. No.: **18/970260**

Filed: **December 05, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20250091683 A1	Mar. 20, 2025

Related U.S. Application Data

continuation parent-doc US 17460515 20210830 US 11753105 20230912 child-doc US 18214769
continuation-in-part parent-doc US 18214769 20230627 US 12162560 child-doc US 18970260

Publication Classification

Int. Cl.: **B62K5/10** (20130101); **B60G17/015** (20060101); **B60G17/016** (20060101); **B62J45/415** (20200101); **B62K5/027** (20130101); **B62K5/05** (20130101); **B62K5/06** (20060101); **B62K5/08** (20060101); **B62K21/00** (20060101)

U.S. Cl.:

CPC **B62K5/10** (20130101); **B60G17/0157** (20130101); **B60G17/0162** (20130101); **B62J45/4151** (20200201); **B62K5/027** (20130101); **B62K5/05** (20130101); **B62K5/06** (20130101); **B62K5/08** (20130101); **B62K21/00** (20130101); B60G2204/4193 (20130101); B60G2204/62 (20130101); B60G2300/122 (20130101); B60G2400/0511 (20130101)

Field of Classification Search

CPC: B60R (21/05); B62D (1/16); B62D (1/19); B62D (7/16)

USPC: 280/777; 280/93.51

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3672697	12/1971	Knowles	74/495	B62D 1/197
3998468	12/1975	Stegmaier	280/777	B62D 1/19
9452806	12/2015	Hirayama	N/A	B62K 21/20
11427249	12/2021	Doerksen	N/A	B60G 17/018

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2017017636	12/2016	WO	B62D 21/11

Primary Examiner: Dickson; Paul N

Assistant Examiner: Miller; Caitlin Anne

Attorney, Agent or Firm: Smith & Hopen, P. A.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This nonprovisional application is a Continuation-in-Part of and claims priority to U.S. Nonprovisional application Ser. No. 18/214,769 entitled “THREE-WHEEL MOTOR VEHICLE AND CONTROL SYSTEM,” filed on Jun. 27, 2023, by the same inventor, which is a continuation of and claims priority to U.S. Nonprovisional application Ser. No. 17/460,515, now U.S. Pat. No. 11,753,105 entitled “THREE-WHEEL MOTOR VEHICLE AND CONTROL SYSTEM,” filed Aug. 30, 2021, by the same inventor, all of which are incorporated herein by reference, in their entireties, for all purposes.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) This invention relates, generally, to vehicles. More specifically, it relates to three-wheel

vehicles.

2. Brief Description of the Prior Art

(2) Three-wheel vehicles (also referred to as “trikes”) are notoriously unstable and have poor turning capabilities. This characteristic is a result of the inability to lean the vehicle into turns. Instead, the centrifugal force imparted on the vehicle while turning into a corner (also referred to as “cornering”) often causes the wheel on the inside of the turn to come off the ground. Obviously, it can be dangerous when a wheel is lifted off the ground during cornering, especially when cornering at a high rate of speed. If the driver is not careful the trike can flip.

(3) Efforts have been made to improve the cornering abilities of trikes. However, these approaches have resulted in only minor improvements. For example, U.S. Pat. No. 10,435,104 to Terracraft Motors Inc. describes a complex system relying on a suite of sensors and a complicated front beam assembly. While Terracraft's trike is adapted to lean into turns, the trike can only up to about 20 degrees. Moreover, the lean is achieved by leaning the front wheels of the trike. This approach is reliant on a front wheel assembly comprised of A-Arms and wheels with a U-shaped contacting surface to allow the wheels to lean and maintain ground contact. Ultimately, the Terracraft trike is overly complex and provides less than optimal lean characteristics.

(4) Accordingly, what is needed is an improved trike design and lean control system. However, in view of the art considered as a whole at the time the present invention was made, it was not obvious to those of ordinary skill in the field of this invention how the shortcomings of the prior art could be overcome.

(5) All referenced publications are incorporated herein by reference in their entirety. Furthermore, where a definition or use of a term in a reference, which is incorporated by reference herein, is inconsistent or contrary to the definition of that term provided herein, the definition of that term provided herein applies and the definition of that term in the reference does not apply.

(6) While certain aspects of conventional technologies have been discussed to facilitate disclosure of the invention, Applicants in no way disclaim these technical aspects, and it is contemplated that the claimed invention may encompass one or more of the conventional technical aspects discussed herein.

(7) The present invention may address one or more of the problems and deficiencies of the prior art discussed above. However, it is contemplated that the invention may prove useful in addressing other problems and deficiencies in a number of technical areas. Therefore, the claimed invention should not necessarily be construed as limited to addressing any of the particular problems or deficiencies discussed herein.

(8) In this specification, where a document, act or item of knowledge is referred to or discussed, this reference or discussion is not an admission that the document, act or item of knowledge or any combination thereof was at the priority date, publicly available, known to the public, part of common general knowledge, or otherwise constitutes prior art under the applicable statutory provisions; or is known to be relevant to an attempt to solve any problem with which this specification is concerned.

BRIEF SUMMARY OF THE INVENTION

(9) The long-standing but heretofore unfulfilled need for an improved trike design and lean control system is now met by a new, useful, and nonobvious invention.

(10) The novel structure of the three wheel motor vehicle includes a chassis having a front end and a rear end with a rotational control shaft proximate the front end of the chassis. The rotational control shaft has a rotational axis that is directionally oriented along a path generally extending from the front end of the chassis to the rear end of the chassis. The chassis also provides structural support for a motor/engine, driver's seat, and vehicle controls.

(11) In some embodiments, the rotational axis of the rotational control shaft is directionally oriented such that the rotational axis intersects a ground surface at or before an area of contact of a rear drive wheel with the ground surface. In some embodiments, the rotational axis of the rotational

control shaft is directionally oriented such that the rotational axis intersects a ground surface aft of an area of contact of a rear drive wheel with the ground surface.
(12) The vehicle further includes a front wheel assembly. The front wheel assembly includes a pair of wheels, a beam extending between the pair of wheels, and a first lean control motor operably coupled to the rotational control shaft of the chassis, such that actuation of the first lean control motor causes rotation of the chassis about the rotational axis of the rotational control shaft.
(13) In some embodiments, the first lean control motor is a rotational motor with a rotational shaft. The rotational shaft of the first lean control motor is mechanically connected to or integrated with the rotational control shaft of the chassis.
(14) Some embodiments further include a worm drive gear assembly structurally supported by the front wheel assembly. The worm drive gear assembly includes a first input shaft operably connected to the first lean control motor and an output shaft operably connected to the rotational control shaft of the chassis. The first input shaft is operably coupled with the output shaft via a worm screw meshed with a worm wheel. Thus, operation of the first lean control motor causes rotation of the first input shaft and in turn causes rotation of the output shaft and the rotational control shaft of the chassis.
(15) In some embodiments, the worm drive gear assembly further includes a second input shaft operably connected to a second lean control motor. The second input shaft is operably coupled with the output shaft via the worm screw meshed with the worm wheel. Thus, operation of the second lean control motor causes rotation of the second input shaft and in turn causes rotation of the output shaft and the rotational control shaft of the chassis. In some embodiments, the gear assembly is self-locking meaning that the worm wheel cannot drive the worm screw, or in other words, the gear assembly prevent back driving.
(16) Some embodiments of the present invention further comprise a lean control system. The lean control system includes a lean indicator sensor and control circuitry. The lean indicator sensor is calibrated to identify a normal plane of the chassis, wherein the normal plane extends from a bottom to a top of the chassis and is perpendicular to a lateral/cross axis of the chassis. The lean indicator sensor is also configured to detect a resultant force on the vehicle that is not parallel to the normal plane. The control circuitry is configured to read the outputs from the lean indicator sensor and control the operation of the first lean control motor. In addition, the control circuitry can determine the direction of the resultant force on the vehicle based on the outputs from the lean indicator sensor. In response to detecting that the resultant force on the vehicle is not parallel to the normal plane, the control circuitry is configured to cause the first rotational motor to rotate the rotational control shaft to rotate the chassis and its normal plane into parallel alignment with the resultant force.
(17) Some embodiments of the present invention further include a cable steering system. The cable steering system has a steering wheel and a steering cable operably connected to the steering wheel and to one or more of the front wheels. Thus, input from the steering wheel causes the one or more front wheels to pivot about a respective king pin. Moreover, the cable steering system isolates steering inputs from the vehicle's leaning capabilities.
(18) In some embodiments, each front wheel is secured to the front wheel assembly by a king pin and the kingpin has a longitudinal axis that is offset between 0 and 5 degrees from a vertical axis. In some embodiments, a suspension system is operably coupled to the front wheel assembly. The suspension system has a shock absorber and preferably a spring with a longitudinal axis that is offset between 20 and 25 degrees from the vertical axis.
(19) In some embodiments, the front wheel assembly includes a first beam and a second beam operably coupled to the two front wheels, with the first beam vertically spaced from the second beam at the attachment point of the beams.
(20) In some embodiments, the front two tires are wider and flatter than the rear drive wheel.
(21) Some embodiments of the present invention include a front wheel assembly for a three wheel

motor vehicle. The front wheel assembly includes a rotational control shaft configured to be mechanically secured to a chassis in a generally non-rotational manner, such that rotation of the rotational control shaft causes rotation of the chassis when the rotational control shaft is secured to the chassis. The rotational control shaft has a rotational axis that is directionally oriented along a path generally extending downwardly towards a ground surface in a rearward direction when the rotational control shaft is secured to the chassis.

(22) The front wheel assembly further includes a first lean control motor operably coupled to the rotational control shaft of the chassis, such that actuation of the first lean control motor causes rotation of the chassis about the rotational axis of the rotational control shaft. In some embodiments, the first lean control motor is a rotational motor with a rotational shaft. The rotational shaft of the first lean control motor is mechanically connected to or integrated with the rotational control shaft.

(23) Some embodiments further include a worm drive gear assembly structurally supported by the front wheel assembly. The worm drive gear assembly includes a first input shaft operably connected to the first lean control motor and an output shaft operably connected to the rotational control shaft. The first input shaft is operably coupled with the output shaft via a worm screw meshed with a worm wheel. Thus, operation of the first lean control motor causes rotation of the first input shaft and in turn causes rotation of the output shaft and the rotational control shaft.

(24) In some embodiments, the worm drive gear assembly further includes a second input shaft operably connected to a second lean control motor. The second input shaft is operably coupled with the output shaft via the worm screw meshed with the worm wheel. Thus, operation of the second lean control motor causes rotation of the second input shaft and in turn causes rotation of the output shaft and the rotational control shaft. In some embodiments, the gear assembly is self-locking meaning that the worm wheel cannot drive the worm screw, or in other words, the gear assembly prevent back driving.

(25) Some embodiments of the present invention further comprise a lean control system. The lean control system includes a lean indicator sensor and control circuitry. The lean indicator sensor is calibrated to identify a normal plane of the chassis, wherein the normal plane extends from a bottom to a top of the chassis and is perpendicular to a lateral/cross axis of the chassis. The lean indicator sensor is also configured to detect a resultant force on the vehicle that is not parallel to the normal plane. The control circuitry is configured to read the outputs from the lean indicator sensor and control the operation of the first lean control motor. In addition, the control circuitry can determine the direction of the resultant force on the vehicle based on the outputs from the lean indicator sensor. In response to detecting that the resultant force on the vehicle is not parallel to the normal plane, the control circuitry is configured to cause the first rotational motor to rotate the rotational control shaft to rotate the chassis and its normal plane into parallel alignment with the resultant force.

(26) The front wheel assembly further includes a beam extending in generally a perpendicular direction with respect to the rotational control shaft. The beam is configured to be operably secured to a pair of front wheels. In some embodiments, the front wheel assembly includes a first beam and a second beam operably coupled to the two front wheels, with the first beam vertically spaced from the second beam at the attachment point of the beams.

(27) In some embodiments, each front wheel is secured to the front wheel assembly by a king pin and the kingpin has a longitudinal axis that is offset between 0 and 5 degrees from a vertical axis.

(28) In some embodiments, a suspension system is operably coupled to the front wheel assembly. The suspension system has a shock absorber and preferably a spring with a longitudinal axis that is offset between 20 and 25 degrees from the vertical axis.

(29) In some embodiments, the front wheel assembly is provided with a cable steering system. The cable steering system has a steering wheel and a steering cable operably connected to the steering wheel and to one or more of the front wheels. Thus, input from the steering wheel causes the one or

more front wheels to pivot about a respective king pin. Moreover, the cable steering system isolates steering inputs from the vehicle's leaning capabilities.
(30) These and other important objects, advantages, and features of the invention will become clear as this disclosure proceeds.
(31) The invention accordingly comprises the features of construction, combination of elements, and arrangement of parts that will be exemplified in the disclosure set forth hereinafter and the scope of the invention will be indicated in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) For a fuller understanding of the invention, reference should be made to the following detailed description, taken in connection with the accompanying drawings, in which:
(2) FIG. **1** is a side view of an embodiment of the present invention.
(3) FIG. **2** is a front view of an embodiment of the present invention.
(4) FIG. **3** is a front view of an embodiment of the present invention when leaning at a 20-degree angle.
(5) FIG. **4** is a front view of an embodiment of the present invention when leaning at a 45-degree angle.
(6) FIG. **5** is a front sectional view of an embodiment of the front end assembly.
(7) FIG. **6** is an overhead sectional view of an embodiment of the front end assembly.
(8) FIG. **7**A is a side view of an embodiment of the present invention.
(9) FIG. **7**B is an overhead view of an embodiment of the present invention.
(10) FIG. **8**A is a side view of an embodiment of the present invention.
(11) FIG. **8**B is an overhead view of an embodiment of the present invention.
(12) FIG. **9**A is a side view of an embodiment of the present invention.
(13) FIG. **9**B is an overhead view of an embodiment of the present invention.
(14) FIG. **10** is a front sectional view of an embodiment of the present invention.
(15) FIG. **11** is an overhead view of an embodiment of a front wheel.
(16) FIG. **12** is a side view of an embodiment of a front wheel depicting caster angles of the suspension system and steering kingpin.
(17) FIG. **13** is a simplified diagram of an embodiment of the steering system.
(18) FIG. **14** is a side view of an embodiment of the present invention.
(19) FIG. **15** is a block diagram of an embodiment of the lean control system.
(20) FIG. **16** is a front view of an embodiment of the front wheel assembly of the present invention from the perspective of the driver's seat.
(21) FIG. **17** is an overhead view of an embodiment of the front wheel assembly of the present invention.
(22) FIG. **18** is an overhead view of an embodiment of the front wheel assembly depicting the wheels turned at an angle of the present invention.
(23) FIG. **19** is a side cross-sectional view of an embodiment of a front wheel of the present invention in a vertical position.
(24) FIG. **20** is a side, cross-section view of an embodiment of a front wheel of the present invention during operation.
(25) FIG. **21**A is an overhead view of an embodiment of the guide tube system of the present invention.
(26) FIG. **21**B is a side view of an embodiment of the guide tube system of the present invention.
(27) FIG. **21**C is a back view of an embodiment of the guide tube system of the present invention.
(28) FIG. **22**A is an overhead view of an embodiment of the tie rod system of the present invention.

(29) FIG. **22**B is a side view of an embodiment of the tie rod system of the present invention.
(30) FIG. **22**C is a back view of an embodiment of the tie rod system of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(31) In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings, which form a part thereof, and within which are shown by way of illustration specific embodiments by which the invention may be practiced. It is to be understood that other embodiments may be utilized, and structural changes may be made without departing from the scope of the invention.
(32) As used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the context clearly dictates otherwise.
(33) In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of embodiments of the present technology. It will be apparent, however, to one skilled in the art that embodiments of the present technology may be practiced without some of these specific details.
(34) The phrases “in some embodiments,” “according to some embodiments,” “in the embodiments shown,” “in other embodiments,” and the like generally mean the particular feature, structure, or characteristic following the phrase is included in at least one implementation. In addition, such phrases do not necessarily refer to the same embodiments or different embodiments.
(35) The term “lateral force” refers to forces that are nonparallel to a vertical reference plane extending through the trike when the trike is stationary and upright. Lateral forces may be centrifugal forces, wind forces, or any other external force acting on the trike. When a lateral force is imparted on the vehicle, the vehicle experiences a resultant force in a direction based on the magnitude of the lateral force relative to the force of gravity.
(36) The “vertical reference axis” is depicted in FIGS. **2-4** using reference numeral **107** and is parallel with the normal plane of the trike when the trike is stationary and in an upright orientation as depicted in FIG. **2**.
(37) The “normal plane” refers to a plane or axis extending from the top to the bottom of the trike. The normal plane is fixed with respect to the trike and rotates with the chassis as the chassis leans from side to side. The normal plane is depicted in FIGS. **2-4** using reference numeral **108**.
(38) While the exemplary figures provided herein do not depict various common features provided on motor vehicles, a person of ordinary skill in the art would understand that such features are included in various embodiments of the present invention. For example, embodiments of the present invention include one or more seats **101** for occupants, an engine/motor **103** for powering drive wheel **158**, and vehicle control systems for controlling the various aspects of the vehicle, including but not limited to, acceleration, braking, and handling, and other common features found on vehicles, including but not limited to lights, mirrors, and turn signals. Furthermore, while the exemplary images depict a single seat trike, some embodiments include additional seats for multiple occupants. Some embodiments include no seats.
(39) A shown in FIGS. **1-4**, the present invention includes three-wheeled vehicle **100** (also referred to herein as a “trike”) having a unique front wheel assembly **102** attached to a unique chassis **104**. Front end **104**A of chassis **104** includes rotational control shaft (“RCS”) **105** (see e.g., FIGS. **5-6**) having a rotational axis that is generally directed in a longitudinal direction, i.e., a direction extending from the front to the back of trike **100**. RCS **105** is integrated with or secured to chassis **104** in a non-rotational manner. In some embodiments, RCS **105** is secured to chassis **104** via a forward structural mount **109** and a rearward structural mount **111**.
(40) RCS **105** is also passes through front wheel assembly **102** in a rotationally free manner, such that RCS **105** can rotate about its rotational axis. Front wheel assembly **102** includes one or more lean control motors **120**, which are operably configured to rotate RCS **105** about its rotational axis

thereby causing chassis **104** to lean from side to side to improve the handling ability of the vehicle.

(41) As depicted in FIGS. **2-4**, trike **100** is configured to operate in a generally vertical orientation and can lean from side to side. As will be explained in greater detail in subsequent paragraphs, some embodiments of the present invention include a lean control system ("LCS") configured to control the degree of rotation and when chassis **104** is rotated.

(42) Front wheel assembly **102** further includes cross frame **110** spanning between front wheels **116**A and **116**B. As depicted, cross frame **110** includes a first beam **112** and a second beam **114**. Some embodiments, however, may employ a single beam or more than two beams.

(43) As best shown in FIGS. **2-4**, in some embodiments, first beam **112** and second beam **114** attach to wheels **116**A and **116**B at different locations to reduce or prevent cross frame **110** from pitching during acceleration and deceleration. The exemplified embodiment includes first beam **112** and second beam **114** attached to suspension system **118**A and **118**B in a generally vertically offset configuration. However, some embodiments may include first beam **112** and second beam **114** operably connected to wheels **116**A and **116**B through alternative components and/or in non-vertical offsets.

(44) Referring now to FIGS. **5-6**, one or more lean control motors **120** are mounted to front wheel assembly **102** via mounts **121**. As previously noted, lean control motors **120** operably engage RCS **105** of chassis **104**. Lean control motors **120** may be any motor known to a person of ordinary skill in the art that is configured or configurable to impart a rotational force, directly or indirectly, onto RCS **105**. In some embodiments, as depicted in FIGS. **5-6**, lean control motors **120** are electric rotational motors having a rotational shaft **122**. A power source (not shown) is operably connected to lean control motors **120** to provide the necessary power for operating said motors.

(45) In some embodiments, one or more lean control motor(s) **120** are attached directly to RCS **105**. In some embodiments, such as the one depicted in FIGS. **5-6**, lean control motors **120** operably engage one or more intermediate components to rotate RCS **105**. The depicted intermediate component is a worm gear assembly **124** mounted to front wheel assembly **102** via mounting supports **125**. Worm gear assembly **124** includes two input shafts **126**A and **126**B operably connected to lean control motors **120**A and **120**B, respectively. Shafts **126**A and **126**B are connected to shafts **122**A and **122**B via mechanical coupler **128**A and **128**B, respectively. As a result, shafts **126**A and **126**B rotate as one with shafts **122**A and **122**B. In some embodiments, shafts **126**A and **126**B are integrated with shafts **122**A and **122**B or attached to each other through other components known to a person of ordinary skill in the art, such that shafts **126**A and **126**B rotate as one with shafts **122**A and **122**B, respectively. In some embodiments, the intermediate components are self-locking, or in other words, back driving is not possible.

(46) Input shafts **126**A and **126**B are operably connected to worm screw **130**. Typically, input shafts **126**A and **126**B are integrated with or directly connected to worm screw **130**, such that rotation of input shafts **126**A and **126**B causes rotation of worm screw **130**. Worm screw **130** includes a helical thread which meshes with a plurality of projections on worm wheel **132**. This meshed connection transfers the rotational force imposed on worm screw **130** to worm wheel **132**.

(47) Worm wheel **132** is operably connected to output shaft **134**. Output shaft **134** is integrated with or directly connected to worm wheel **132**, such that rotation of worm wheel **132** causes rotation of output shaft **134**.

(48) Output shaft **134** is also operably connected to RCS **105**. The depicted embodiments include a key/key slot connection **136**. However, output shaft **134** may be mechanically connected to RCS **105** through a coupler, integrated with, or directly connected to RCS **105**. Because of the interconnection of output shaft **134** and RCS **105**, rotation of output shaft **134** causes rotation of RCS **134**, which in turn causes rotation of chassis **104**.

(49) Some embodiments use alternative lean control motors and intermediate components known to a person of ordinary skill in the art to convert the output of the one or more lean control motors into rotation of RCS **105**. Embodiments may also use alternative types of gears or other types of

force transferring components that are in mechanical communication with RCS **105** to cause rotation of RCS **105** about its rotational axis.

(50) Referring now to FIGS. **7-9**, embodiments of the present invention include rotational axis **150** of RCS **105** having a predetermined downward angle towards rear end **104**B of chassis **104**, such that the projection of rotational axis **150** has an intersection point **152** with ground **154**. In some embodiments, intersection point **152** of rotational axis **150** is within contact area **156** of rear wheel **158** with ground **154**. In some embodiments, as depicted in FIGS. **7**, intersection point **152** of rotational axis **150** is at center point **160** of contact area **156** of rear wheel **158** with ground **154**. This particular configuration eliminates or minimizes a turning moment on front wheel assembly **102**, such that front wheel assembly **102** maintains a generally perpendicular orientation with chassis **104** when chassis **104** leans to a side. Moreover, this configuration ensures that trike **100** does not oversteer or understeer when chassis **104** is leaning during a turn.

(51) In some embodiments, as shown in FIGS. **8**, intersection point **152** of rotational axis **150** is located fore of contact area **156** of rear wheel **158** with ground **154**. This particular configuration produces a turning moment force on front wheel assembly **102** when chassis **104** leans to a particular side. If chassis **104** leans to the port side, then a moment force is imposed on front wheel assembly **102** attempting to turn front wheel assembly **102** towards the port side of vehicle **100**. As a result, this particular configuration causes trike **100** to have some oversteering characteristics when chassis **104** is leaning during a turn. FIG. **8**B is a simplified illustration depicting how front wheel assembly **102** turns to the port side of trike **100** when chassis **104** initiates a port side lean. It should be noted that front wheel assembly **102** depicted in FIG. **8** is a simplified illustration to clarify how front wheel assembly **102** rotates in this configuration.

(52) In some embodiments, as shown in FIGS. **9**, intersection point **152** of rotational axis **150** is located aft of contact area **156** of rear wheel **158** with ground **154**. This particular configuration produces a turning moment force on front wheel assembly **102** opposite to the direction in which chassis **104** leans. In other words, if chassis **104** leans to the port side, then front wheel assembly **102** experiences a turning moment force on front wheel assembly **102** attempting to turn front wheel assembly **102** towards the starboard side of vehicle **100**. As a result, this particular configuration causes trike **100** to have some understeering characteristics when chassis **104** is leaning during a turn. FIG. **9**B illustrates how front wheel assembly **102** rotates to the starboard side of trike **100** when chassis **104** initiates a port side lean. It should be noted that front wheel assembly **102** depicted in FIG. **9** is a simplified illustration to clarify how front wheel assembly **102** rotates in this configuration.

(53) In some embodiments, RCS **105** is connected to chassis **104** in a pitch adjustable manner. In other words, rotational axis **150** of RCS **105** has an adjustable downward angle towards rear end **104**B of chassis **104**. Some embodiments include a pitch adjusting actuator to alter the pitch angle. In some embodiments a user can alter the pitch angle of RCS **105** based on user input/instruction. As a result, the vehicle can be tuned to have different steering capabilities based on the user's preference. In some embodiments, the pitch adjusting actuator can be used dynamically during operation of the trike to create a turning moment on the trike without using the steering wheel.

(54) Referring now to FIGS. **10-11** some embodiments of front wheel assembly **102** include suspension systems **118**A and **118**B at both front wheels **116**A and **116**B. Each suspension system **118** includes shock absorber **162**, which may be any device, including but not limited to a mechanical or hydraulic device, designed to absorb and dampen shock impulses passing through the front wheels. Embodiments of suspension systems **118** also include springs that work in conjunction with shock absorbers **162**.

(55) Beams **112** and **114** are secured or integrated with translatable tube **164** in vertically spaced relation. Thus, beams **112** and **114** will translate in a generally vertical direction as shock absorbers **162** absorb forces. In some embodiments, the stroke length of the shock absorbers **162** is about 3 inches to about 24 inches.

(56) While shock absorbers **162** allow for independent shock compression between the two front wheels, front wheel assembly **102** will generally maintain a parallel orientation with the ground surface regardless of how chassis **104** is leaning. Generally, the only tilt/rotation about the longitudinal axis of trike **100** of front wheel assembly **102** is contributed to the vertical movement of shock absorbers **162**. Thus, during operation of trike **100**, front wheels **116** operate more similar to car wheels than motorcycle wheels. In contrast, the ability of chassis **104** to lean results in rear wheel **158** leaning as well. Therefore, some embodiments of the present invention, as best depicted in FIGS. **2-4**, use front wheels **116** that are generally flatter and wider than rear wheel **158**. Rear wheel **158** has more of a U-shaped profile similar to motorcycle tires and front tires **116** have a profile shape more similar to tires meant for cars and trucks.
(57) In some embodiments, shock absorbers **162** are vertically offset by a caster angle. This feature is exemplified in the simplified illustration of FIG. **12**. In some embodiments, the caster angle of shock absorbers **162** is a positive caster. In some embodiments, the caster angle of shock absorbers **162** is between 15 and 20 degrees. In some embodiments, the caster angle of shock absorbers **162** is between about (i.e., +/−10%) 20 degrees. In some embodiments, the caster angle of shock absorbers **162** is adjustable to vary the suspension characteristics.
(58) In some embodiments, as depicted in FIGS. **11** and **12**, each suspension system **118** is hingedly connected to its respective front wheel **116** via steering kingpin **164**. Steering kingpin **164** provides the connection to which suspension system **118** is mechanically secured to steering control arm **166** and the pivot about which steering system **170** can control the orientation of front wheels **116** (see FIG. **13**).
(59) In some embodiments, steering kingpins **164** are vertically offset by a caster angle. This feature is also exemplified in the simplified illustration of FIG. **12**. In some embodiments, the caster angle of steering kingpins **164** is a positive caster. In some embodiments, the caster angle of steering kingpins **164** is between 0 and 13 degrees. In some embodiments, the caster angle of steering kingpins **164** is between 0 and 5 degrees. In some embodiments, the caster angle of steering kingpins **164** is between about (i.e., +/−10%) 3 degrees. In some embodiments, the caster angle of steering kingpins **164** is adjustable to vary the steering characteristics. In some embodiments, the caster angle of steering kingpins **164** is distinct from the caster angle of shock absorbers **162**.
(60) Some embodiments of the present invention also include cable steering system **170** as exemplified in FIG. **13**. Steering system **170** is connected to steering wheel **172** via steering column **174**, which passes through dash **176**. Steering column **174** is operably connected to worm gearbox **178** housing worm gear assembly **181**. The worm gear assembly **181** is operably connected to a pair of shank balls **179**, which are operably connected to cables **180**. Cables **180** are in turn secured to steering tie rod **182**, which is operably connected to steering control arms **166**. This particular steering system allows the cable steering system **170** to isolate steering inputs from the vehicle's leaning capabilities. Thus, there is no relationship between the steering system/inputs and the lean of the chassis.
(61) As provided in FIGS. **14** and **15**, some embodiments of the present invention include lean control system (LCS) **106** configured to automatically lean/rotate chassis **104** in response to a lateral force imposed on chassis **104**. LCS **106** can be embodied as a computer device, special-purpose hardware (e.g., circuitry), as programmable circuitry appropriately programmed with software and/or firmware, or as a combination of special-purpose and programmable circuitry. Hence, embodiments may include a machine-readable medium having stored thereon instructions which may be used to program a computer (or other electronic devices) to perform a process. The machine-readable medium may include, but is not limited to, floppy diskettes, optical disks, compacts disc read-only memories (CD-ROMs), magneto-optical disks, ROMs, random access memories (RAMs), erasable programmable read-only memories (EPROMs), electrically erasable programmable read-only memories (EEPROMs), magnetic or optical cards, flash memory, or other

type of media/machine-readable medium suitable for storing electronic instructions.
(62) LCS **106** is in direct or indirect communication (wired or wireless) with lean control motor(s) **120** and lean indicator sensor **182**. More specifically, LCS **106** receives information from lean indicator sensor **182** regarding the forces imparted on trike **100** and directs lean control motor(s) **120** to lean chassis **104** as needed to improve the handling capabilities of trike **100**.
(63) LCS **106** may control a single motor **120**, two lean control motors **120**, or a plurality of lean control motors **120**. When a single lean control motor is used, LCS **106** causes lean control motor **120** to rotate RCS **105** in a predetermined direction based on the readings received from lean indicator sensor **182**. LCS **106** causes lean control motor **120** to rotate RCS **105** in the opposite direction based on the readings received from lean indicator sensor **182**. If multiple lean control motors are used, LCS **106** can use more than one motor to rotate RCS **105** in either direction, can use one motor to slow or stop rotation of RCS **105**, and/or can use one motor to reverse rotation of RCS **105**.
(64) Lean indicator sensor **182** includes one or more sensors configured to detect lateral forces and/or the resultant force on trike **100**. A resultant force is the combination of the force of gravity with one or more additional forces imposed on trike **100**. The resultant force is equal to the gravitational force when no additional forces are imposed on trike **100**. When a lateral force is imposed on trike **100**, the direction of the resultant force is shifted out of alignment with the direction of gravity.
(65) Sensor system **182** may include an inclinometer, tilt indicator, tilt sensor, tilt meter, slope alert, slope gauge, gradient meter, gradiometer, level gauge, level meter, declinometer, roll indicator, or any other sensor known to a person of ordinary skill in the art to detect the resultant force on trike **100**. In some embodiments, sensor system **182** is also configured to identify the direction of the resultant force with respect to normal plane **108** of chassis **104** and convey that information to LCS **106**.
(66) For example, some embodiments of lean indicator sensor **182** are configured to output a voltage to LCS **106** between a predetermined voltage range based on the readings of the one or more sensors. In an embodiment a sensor has an output voltage range from 0 to 5 volts with 2.5 volts corresponding to the resultant force aligned with normal plane **108**. If lean indicator sensor **182** detects a resultant force angled towards, for example the port side of trike **100**, an output voltage between 0 and 2.5 is sent to LCS **106**, where the particular voltage value is based on the angle of the resultant force relative to normal plane **108**. If lean indicator sensor **182** detects a resultant force angled towards the opposite side (starboard side in this example) of trike **100**, an output voltage between 2.5 and 5 is sent to LCS **106**, where the particular voltage value is based on the angle of the resultant force relative to normal plane **108**.
(67) LCS **106** is configured to associate the received voltage from lean indicator sensor **182** with a particular degree of rotation of RCS **105**. Since RCS **105** is controlled via lean control motors **120**, LCS **106** is configured to associate the received voltage from lean indicator sensor **182** with a particular degree of rotation of one or more lean control motors **120**. Moreover, LCS **106** is configured to have a predetermined power curve for operation of the one or more lean control motors **120** to ensure a steady rotation to various degrees of rotation.
(68) Some embodiments also include lean limiters configured to stop the rotation of RCS **105** at a predetermined degree of rotation. The lean limiters ensure that chassis **104** does not rotate to a degree in which rear wheel **158** loses traction with the ground. In some embodiments, the lean limiters prevent rotation beyond 45 degrees in the clockwise and 45 degrees in the counterclockwise direction. In other words, lean limiters restrict the total rotation, from one stop to the other, to a combined 90 degrees of rotation. In some embodiments, the lean limiters prevent rotation beyond 38 degrees in a clockwise and 38 degrees in a counterclockwise direction. In some embodiments, the lean limiters are adjustable to account for different rear wheel sizes and shapes.
(69) In some embodiments, lean limiters are mechanical stops on RCS **105**, worm gear assembly

124, and/or lean control motors **120**. In some embodiments, the lean limiters are software/circuitry based controls. LCS **106** can be configured to identify the degree of rotation of RCS **105** with respect to vertical reference axis **107**, normal plane **108**, and/or the combination of the two and stop rotation at the predefined rotational limits. LCS **106** may also rely on input from the various sensors described herein to determine whether RCS **105** has reached the predetermined rotation limits.

(70) During operation, LCS **106** is constantly adjusting RCS **105** based on the readings from lean indicator sensor **182**. When trike **100** is stationary or when traveling in a straight line without any external lateral forces, LCS **106** keeps chassis **104** in an upright vertical orientation as exemplified in FIG. **2**. When trike **100** is subject to centrifugal forces or any other lateral/non-vertical force, LCS **106** will detect the direction of the resultant force on trike **100** and rotate chassis **104** into the force as exemplified in FIGS. **3** and **4**.

(71) In some embodiments, LCS **106** rotates chassis **104** to align normal plane **108** with the direction of the detected force imparted on chassis **104**. Thus, as shown in in FIG. **3**, when LCS **106** detects a resultant force imposed on trike **100** at a 20-degree angle from the vertical reference axis **107** or normal plane **108**, LCS **106** causes lean control motor(s) **120** to rotate RCS **105** and in turn chassis **104** to align normal plane **108** with the 20-degree angle of the resultant force. As shown in FIG. **4**, when the angle of the detected resultant force on chassis **104** is greater than or equal to 45 degrees (assuming 45 degrees from vertical reference axis **107** is the lean limit), chassis **104** is rotated to bring reference plane **108** to a 45-degree rotation with respect to vertical reference axis **107**. LCS **106** continues to monitor the angle of the resultant force and causes rotation of RCS **105** to align normal plane **108** with the angle of the resultant force on trike **100**.

(72) Some embodiments include feedback sensor **184** in communication with LCS **106**. Feedback sensor **184** is configured to determine whether RCS **105** has been properly rotated to align normal plane **108** with the angle of the resultant force on trike **100**. In some embodiments, feedback sensor **184** is secured to front wheel assembly **102** and operably engages RCS **105** to determine the degree of rotation of RCS **105**. In some embodiments, feedback sensor **184** is secured to front wheel assembly **102** and operably engages chassis **104** and/or RCS **105** to determine the degree of rotation of chassis **104**/RCS **105**.

(73) In some embodiments, feedback sensor **184** is a rotary sensor configured to detect the current rotational position of chassis **104**/RCS **105**. Some embodiments use a linear sensor or potentiometer. Alternative sensors known to a person of ordinary skill can be used as feedback sensor **184**.

(74) Some embodiments include a ground slope sensor in communication with LCS **106**. The ground slope sensor is configured to determine when front wheel assembly **102** rotates with respect to vertical reference plane **107** as a result of the slope/camber of the ground. The ground slope sensor conveys the angle of rotation/slope of front wheel assembly **102** to LCS **106** and LCS **106** factors this rotation into the rotation of RCS **105** and in turn chassis **104**.

(75) As shown in FIGS. **16-22**C, an additional aspect of the present invention pertains to mitigating bump steer during travel and/or the effect of unintended changes in a toe angle of a vehicle as the suspension of the vehicle moves through its range of motion. In some embodiments, the assembly to neutralize bump steer for a wheel assembly on a motor vehicle may comprise a tie rod system. Additionally, in some embodiments, the assembly to neutralize bump may further comprise a guide tube system.

(76) Moreover, in some embodiments, the tie rod system includes a tie rod that can extend between a pair of wheels in the wheel assembly. The tie rod may have a first end and a second end opposing each other. Furthermore, in some embodiments, the first end and second end of the tie rod are operably coupled to a guide tube system of each wheel.

(77) Additionally, the tie rod system may comprise a tie rod shelf. The tie rod shelf may be operably coupled to a beam disposed between the pair of wheels such that at least one portion of

the tie rod can be configured to translate about a surface of the rod shelf.

(78) Furthermore, in some embodiments, the tie rod system may include a cap. The cap can be configured to operably secure the tie rod while in operable communication with the tie rod shelf such that the cap may prevent independent vertical translation of the tie rod.

(79) Moreover, in some embodiments, the guide tube system may comprise a guide structure and a translation member. The guide structure is depicted in the exemplary figures in the form of guide rod **402** but it should be understood that guide structure embodies any other structure that allows for the translation member about a body of the guide structure. The translation member is depicted in the exemplary figures in the form of guide tube **404** but it should be understood that translation member embodies any other structure that can translate about a body of the guide structure.

(80) Finally, in some embodiments, the first end and second end of the tie rod can be operably coupled to the translation member such that the tie rod system can translate a vertical length along the guide structure.

(81) Referring now to FIG. **16**, a front view of an embodiment of the front wheel assembly **102** is depicted incorporating an assembly to neutralize bump steer for a wheel assembly on a motor vehicle. In some embodiments, the assembly to neutralize bump steer comprises a tie rod system **300**. Tie rod system **300** may comprise of a tie rod **302** that may extend between the pair of wheels of the front wheel assembly **102**. In some embodiments, tie rod **302** may be cambered such that worm gearbox **178** may be positioned above tie rod **302**. Alternatively, the camber may allow worm gearbox **178** to be positioned below tie rod **302**.

(82) Additionally, as shown in FIG. **16** in conjunction with FIGS. **17-18**, in some embodiments, tie rod **302** compromises of a first end and second end of tie rod **302**. The first end and second end of tie rod **302** may be operably coupled to the guide tube system **400**. In some embodiments, the first end and second end of tie rod **302** are oppositely threaded from each other. However, in some alternative embodiments, tie rod **302** can operably engage with guide tube system **400** through other known mechanisms or methods, including but not limited to magnets, clamps, screws, crimping, welding, latches, soldering, and/or brazing.

(83) As shown in FIG. **16**, in conjunction with FIGS. **22**A-**22**C, in some embodiments, tie rod system **300** may comprise of a tie rod shelf **310**. Tie rod shelf **310** may be coupled to at least one surface of first beam **112** and/or second beam **114**. For ease of reference, the exemplary embodiment disclosed herein refers to welding as the method of coupling tie rod shelf **310** to second beam **114**, but this description should not be interpreted as exclusionary to other methods of coupling.

(84) Furthermore, in some embodiments, tie rod shelf **310** can be in the form of an elongated support structure establishing a flat and/or shelf-like surface. This surface provides a support upon which tie rod **302** may rest upon the top surface of tie rod shelf **310**.

(85) Moreover, in some embodiments, tie rod system **300** can include a cap **304**. Cap **304** can be configured to operably secure tie rod **302** such that cap **304** may be in operable communication with tie rod shelf **310**. Cap **304** being in operable communication with tie rod shelf **310** prevents independent vertical translation of tie rod **302**. Non-limiting examples of mechanisms and methods of securing cap **304** to tie rod shelf **310** include magnets, snap fasteners, clamps, screws, and/or bolts. For ease of reference, the exemplary embodiment disclosed herein employs bolts, but this description should not be interpreted as exclusionary to other methods of having cap **304** in operable communication with tie rod shelf **310**.

(86) As shown in FIG. **22**B, in some embodiments, cap **304** may be configured to allow for translation of tie rod **302** across a surface of tie rod shelf **310** as indicated by directional arrows **194**. Moreover, in some embodiments, cap **304** and tie rod shelf **310** may be in operable communication such that there may be sufficient clearance to allow translation of tie rod **302** across a surface of tie rod shelf **310** with minimal friction and restrict unnecessary independent vertical translation of tie rod **302**. Additionally, in some embodiments, as shown in FIG. **22**A and FIG. **22**C,

tie rod **302** can translate in a direction perpendicular to the central axis of tie rod shelf **310** as indicated by directional arrows **193**.

(87) In some embodiments, cap **304** may extend in a longitudinal direction relative to the surface of tie rod shelf **310** such that tie rod **302** may freely translate horizontally along a surface of tie rod shelf **310** to a predetermined length during turning of front wheel assembly. In addition, cap **304** may be in the form of any shape or body that can prevent independent vertical translation of tie rod **302**.

(88) Additionally, as shown in FIG. **16** in conjunction with FIG. **22**B, in some alternative embodiments, the tie rod system may comprise of a plurality of tie rod shelves **310** which may be mechanically coupled along first beam **112** and/or second beam **114**. A plurality of caps **304** may each be temporarily affixed to a surface of each tie rod shelves **310** in the same manner previously described. The increased number of tie rod shelf **310** and caps **304** to tie rod system **300** assist in preventing tie rod **302** from moving in a vertical direction independent of wheel assembly **102** such that the wheel system moves the same length in a vertical direction as the tie rod system **302**.

(89) As shown in FIG. **16**, in conjunction with FIGS. **21**A-**21**C, the assembly to neutralize bump steer for a wheel assembly may further comprise a guide tube system **400**. Each wheel of front wheel assembly **102** may have guide tube system **400** coupled to steering control arm **166** of each wheel.

(90) As shown in FIGS. **21**A-**22**C, in an embodiment, guide tube system **400** may comprise of a guide rod **404**. Guide rod **404** may be operably coupled to steering control arm **166**. Guide tube system **400** may further comprise of a guide tube **402**. Guide tube **402** may slide along the longitudinal length of guide rod **404**.

(91) In some embodiments, guide tube **402** may comprise of an aperture through the central axis of guide tube **402** such that the diameter of the aperture can be greater than the diameter of guide rod **404** allowing guide tube **402** to slide along guide rod **404**. In some alternative embodiments, guide tube **402** may take the form of any member that may allow for translation along the length of guide rod **404**.

(92) Moreover, in some embodiments, as shown in FIGS. **21**A-**21**C, guide tube **402** may radially pivot 360 degrees about the longitudinal axis of guide rod **404** as shown by directional arrows **196**. In some alternative embodiments, guide tube **402** may further comprise linear bearings and/or bushings along the inner surface of guide tube **402** to reduce friction between guide tube **402** and guide rod **404** as guide tube **402** rotates and translates in a vertical direction about guide rod **404**.

(93) Furthermore, as shown in FIGS. **21**A-**21**C, in some embodiments, guide tube system **400** may further comprise of a yoke **406**. Yoke **406** being mechanically coupled to the surface of both guide tube **402** and tie rod **302**. Yoke **406** may be mechanically coupled to the surface of guide tube **403** such that yoke **406** may pivot tie rod **302** about the central axis of guide rod **404** as depicted by arrows **196**.

(94) Additionally, in some alternative embodiments, yoke **406** may comprise of a threaded engagement area for a threaded first end and/or second end of tie rod **302**. In some embodiments, the depth of the threaded engagement area of yoke **406** may be at least three times the diameter of tie rod **302** such that the engagement area can provide additional space to allow tie rod **302** to freely rotate to a predetermined degree of within the engagement area of yoke **406**. In this manner, the ability of tie rod **302** to freely rotate within the engagement area of yoke **406** may reduce the wearing and/or shearing effect a force imparted onto the tie rod may have during steering, suspension, compression, and/or hard braking of trike **100**.

(95) As shown in FIG. **17**, in some embodiments, steering control arm **166** of each wheel of the front wheel assembly **102** is offset from the center of the wheel. In some alternative embodiments, the length and size of steering control arm **166** may be increased to extend farther inward towards the center of trike **100**. As the size of steering control arm **166** is increased, the length of tie rod **302** may be reduced.

(96) In some alternative embodiments, a longer steering control arm **166** for each wheel may be coupled to tie rod **302** such that tie rod **302** need not take a chambered form. In this alternative embodiment, a longer steering control arm **166** for each wheel and a straight tie rod **302** may allow for increased steering ability and decrease in backlash during operation of trike **100**.
(97) Furthermore, as shown in FIG. **17**, in some embodiments, steering control arm **166** can rotate about the radial axis of steering kingpin **164** to a predetermined arc length. Steering control arm **166** can rotate about steering kingpin **164** to range of degrees θ which directly corresponds to the predetermined arc length. As shown in FIGS. **17-18**, predetermined range of degrees θ which steering control arm **166** may rotate about kingpin **164** also corresponds directly to the degree a wheel of front wheel assembly **102** has been turned. The degree of which steering control arm **166** may rotate about steering kingpin **164** may comprise a range of degrees of about 0 degrees to 80 degrees. For example, in an embodiment, steering control arm **166** may rotate from a rest position to a degree of about 62 degrees. In some alternative embodiments, the maximum degree to which steering control arm may rotate about steering kingpin is the final Ackerman position according to the maximum degree of turn of front wheel assembly **102**.
(98) FIG. **18** depicts front wheel assembly **102** rotating to the starboard side of trike **100** when chassis **104** initiates a port side lean. It should be noted that front wheel assembly **102** depicted in FIG. **18** is a simplified illustration to clarify how front wheel assembly **102** rotates in this configuration. As shown in FIG. **18**, in an embodiment, as front wheel assembly **102** rotates to the starboard side of trike **100**, tie rod **302** translates horizontally to a predetermined length along a surface of tie rod shelf **310** due to the rotation of steering control arm **166**.
(99) FIG. **19**, in conjunction with FIG. **20**, depicts either suspension system **118**A or **118**B in a vertical position such that guide tube system **400** is permanently parallel to suspension systems **118**A and **118**B. The parallel orientation of guide tube system **400** reduces the effect of bump steer. Moreover, in some embodiments, tie rod **302** may translate along a perpendicular path to the vertical axis of guide tube system **400** and suspension system **118**A and **118**B.
(100) As shown in FIG. **20**, in some embodiments, suspension system **118**A and **118**B and guide tube system **400** may be positioned in a generally vertically offset configuration. Suspension systems **118**A and **118**B can both be parallel to guide tube system **400** of each wheel. In some embodiments, tie rod **302** may translate in a generally horizontal path which is perpendicular to the generally vertically offset configuration of suspension system **118**A and **118**B and guide tube system **400**.
(101) Furthermore, as shown in FIG. **20**, in conjunction with FIG. **21**A, in some embodiments, steering bracket arm **166** may be mechanically coupled to a caliper plate **192**. In some alternative embodiments, caliper plate **192** may be mechanically coupled to steering kingpin **164** such that steering bracket arm **166** may be offset from steering kingpin **164** allowing for greater rotation.
(102) As shown in FIGS. **16-18** in conjunction with FIG. **19**A, in some embodiments, tie rod system **300** and guide tube system **400** further mitigate bump steer by having steering bracket arm **166** move in a relatively vertical direction. Steering bracket arm **166** may move in a vertical direction directly proportional to suspension system **118** such that the steering bracket arm translates the same vertical length as suspension system **118** and front wheel assembly **102**. Furthermore, tie rod system **300** and guide tube system **400** allow steering bracket arm **166** to translate along a horizontal plane to a predetermined length such that during movement and/or turning of the wheel assembly steering bracket arm **166** may translate along the horizontal plane in the direction front wheel assembly **102** is turning during turning of the motor vehicle. Moreover, in some embodiments, tie rod system **300** and guide tube system **400** allow for steering bracket arm **166** to rotate about the longitudinal axis of kingpin **164** to a predetermined degree as depicted by θ such that the ability for steering bracket arm **166** to rotate about the longitudinal axis of kingpin **164** further mitigates bump steer in front wheel assembly **102**. In some embodiments, steering bracket arm **166** being configured to rotate about the longitudinal axis of kingpin **164** can allow

steering bracket arm **166** to easily turn with front wheel assembly **102** during turning of the motor vehicle further mitigating bump steer.

(103) The advantages set forth above, and those made apparent from the foregoing description, are efficiently attained. Since certain changes may be made in the above construction without departing from the scope of the invention, it is intended that all matters contained in the foregoing description or shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

(104) It is also to be understood that the following claims are intended to cover all of the generic and specific features of the invention herein described, and all statements of the scope of the invention that, as a matter of language, might be said to fall therebetween.

Claims

1. An assembly to neutralize bump steer for a wheel assembly on a motor vehicle, comprising: a tie rod system; a guide tube system; the tie rod system comprising: a tie rod extending between a pair of wheels in the wheel assembly having a first end and second end opposing each other, wherein the opposing ends of the tie rod are operably coupled to a guide tube system of each wheel; a tie rod shelf, the tie rod shelf operably coupled to a beam disposed between the pair of wheels wherein at least one portion of the tie rod is configured to translate about a surface of the rod shelf; a cap configured to operably secure the tie rod in operable communication with the tie rod shelf wherein the cap is intended to prevent independent vertical translation of the tie rod; the guide tube system of each wheel, the guide tube system comprising: a guide structure; a translation member, configured to translate relative to the guide structure; and wherein the first end and second end of the tie rod are operably coupled to the translation member, whereby the tie rod system can translate a vertical length along the guide structure.

2. The assembly to neutralize bump steer for a motor vehicle of claim 1, wherein each guide structure is affixed to a steering control arm of each wheel, the steering control arm configured to pivot about a kingpin of each wheel.

3. The assembly to neutralize bump steer for a motor vehicle of claim 2, further comprising a suspension system operably coupled to the kingpin of each tire.

4. The assembly to neutralize bump steer for a motor vehicle of claim 3, wherein the suspension system has a shock absorber with a longitudinal axis that is offset between 20 and 25 degrees from the vertical axis.

5. The assembly to neutralize bump steer for a motor vehicle of claim 4, wherein the guide tube system is parallel to the suspension system.

6. The assembly to neutralize bump steer for a motor vehicle of claim 1, wherein the guide tube system prevents independent lateral or angular deviations of the tie rod during suspension, compression, or rebound.

7. The assembly to neutralize bump steer for a motor vehicle of claim 1, wherein the guide tube system further comprises a yoke mechanically coupled to both the translation member and one of the opposing ends of the tie rod.

8. The assembly to neutralize bump steer for a motor vehicle of claim 7, wherein the yoke of the guide tube system radially pivots the tie rod system about the guide structure.

9. The assembly to neutralize bump steer for a motor vehicle of claim 1, wherein the length of the tie rod is cambered.

10. The assembly to neutralize bump steer for a motor vehicle of claim 1, wherein the first end and the second end of the tie rod are oppositely threaded whereby subsequent to a force imparted on the tie rod the opposite threading causes the tie rod to freely rotate to a predetermined degree within the yokes of the guide tube system.

11. The assembly to neutralize bump steer for a motor vehicle of claim 10, wherein the depth of the

thread engagement of the yoke is at least three times the diameter of the tie rod.
12. A front wheel assembly for a motor vehicle, comprising: a pair of wheels; a support structure extending between the pair of wheels; assembly to neutralize bump steer extending between the pair of wheels, the assembly to neutralize bump steer comprising: a tie rod system extending between the pair of wheels configured to translate between a fore and aft location of the front wheel assembly; and a guide tube system configured to translate in a generally vertical orientation.
13. The front wheel assembly of claim 12, further comprising a suspension system operably coupled to each wheel, wherein the suspension system has a shock absorber with a longitudinal axis that is offset between 20 and 25 degrees from the vertical axis.
14. The front wheel assembly for a motor vehicle of claim 12, wherein the bump steer assembly comprising: the tie rod system comprising: a tie rod extending between a pair of wheels in the wheel assembly having a first end and second end opposing each other, wherein the opposing ends of the tie rod are operably coupled to a guide tube system of each wheel; a tie rod shelf, the tie rod shelf operably coupled to a beam disposed between the pair of wheels wherein at least one portion of the tie rod is configured to translate about a surface of the rod shelf; a cap configured to operably secure a tie rod in operable communication with the tie rod shelf wherein the cap is intended to prevent independent vertical translation of the tie rod; the guide tube system of each wheel, the guide tube system comprising: a guide structure; and a translation member, configured to translate relative to the guide structure.
15. The assembly to neutralize bump steer for a motor vehicle of claim 14, wherein the guide tube system is parallel to the suspension system.
16. The front wheel assembly for a motor vehicle of claim 14, wherein the guide tube system comprising: a yoke mechanically coupled to both the translation member and the tie rod; and wherein the yoke of the guide tube system radially pivots the tie rod system about the guide structure.
17. A front wheel assembly for a motor vehicle, comprising: a pair of wheels; a support structure extending between the pair of wheels; assembly to neutralize bump steer extending between the pair of wheels, the assembly to neutralize bump steer comprising: a tie rod system extending between the pair of wheels configured to translate between a fore and aft location of the front wheel assembly; and a guide tube system coupled to each wheel of the pair of wheels configured to translate in a generally vertical orientation; wherein the tie rod system is operably coupled to the guide tube system of each wheel; and a first lean control motor operably coupled to the rotational control shaft of the chassis, such that actuation of the first lean control motor causes rotation of the chassis about the rotational axis of the rotational control shaft.
18. The front wheel assembly for a motor vehicle of claim 17, wherein the tie rod system comprising: a tie rod extending between a pair of wheels in the wheel assembly having a first end and second end opposing each other, wherein the opposing ends of the tie rod are operably coupled to a guide tube system of each wheel; a tie rod shelf, the tie rod shelf operably coupled to a beam disposed between the pair of wheels wherein at least one portion of the tie rod is configured to translate about a surface of the rod shelf; and a cap configured to operably secure a tie rod in operable communication with the tie rod shelf wherein the cap is intended to prevent independent vertical translation of the tie rod.
19. The front wheel assembly for a motor vehicle of claim 17, wherein the guide tube system comprising: a guide structure; and a translation member, configured to translate relative to the guide structure.
20. The front wheel assembly for a motor vehicle of claim 18, wherein the guide tube system comprising: a yoke mechanically coupled to both the translation member and the tie rod; and wherein the yoke of the guide tube system radially pivots the tie rod system about the guide structure.